Exp No:1.a Analyze the trend of data science job postings over the last decade

Description: Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather
data on the number of data science job postings each year. Use pandas for data manipulation
and matplotlib/seaborn for visualization.

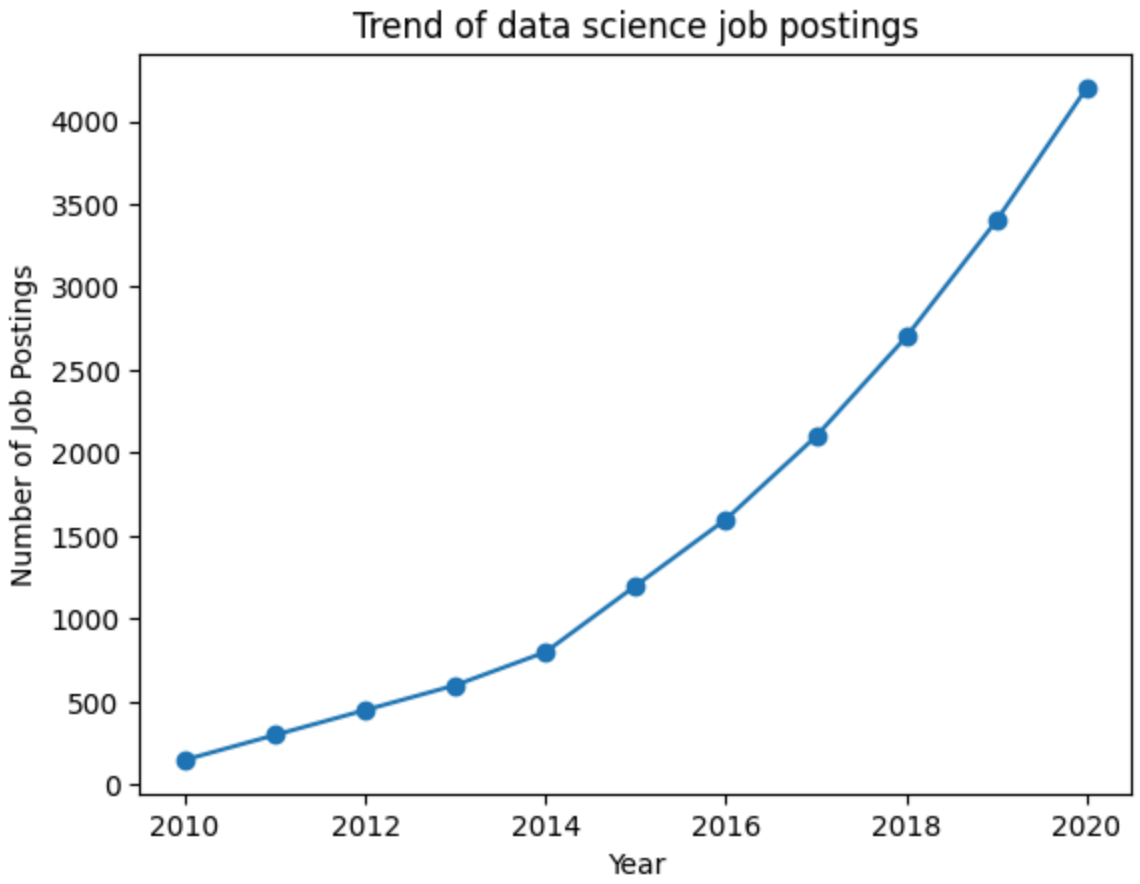
CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    "Year" : list(range(2010,2021)),
    "Job Postings":  [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700,
3400, 4200]
}
df = pd.DataFrame(data)
plt.plot(df["Year"],df["Job Postings"], marker = "o")
plt.title("Trend of data science job postings")
plt.xlabel('Year')
plt.ylabel("Number of Job Postings")
plt.show()
```

Sample Data Input:

Year =2010, 2021
Job Postings=150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200

SAMPLE OUTPUT:



RESULT:

The trend of latest data science job posting is analyzed successfully using line graph.

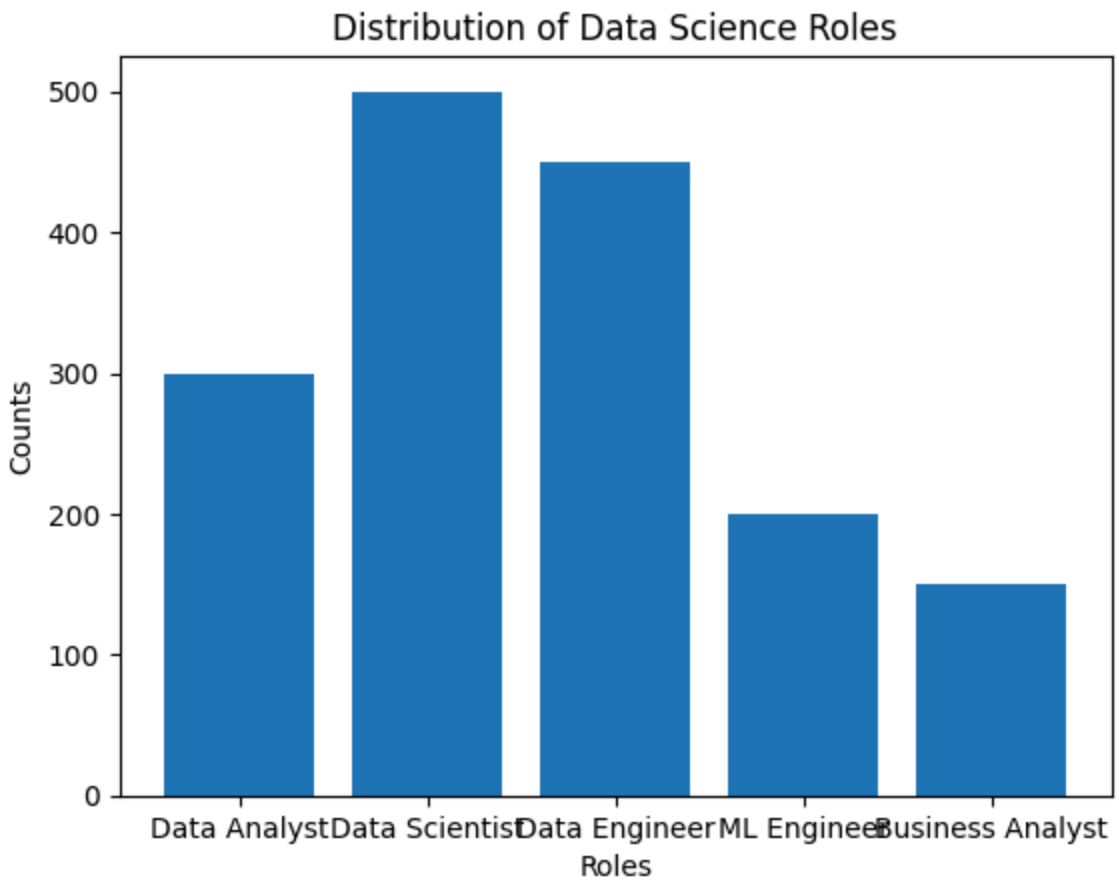
Exp No:1.b Analyze and visualize the distribution of various data science roles (DataAnalyst, Data Engineer, Data Scientist, etc.) from a dataset.

Description: Use a dataset of job postings and categorize them into different roles. Visualize the distribution using pie charts or bar plots.

CODE:

```
import matplotlib.pyplot as plt
roles = ["Data Analyst","Data Scientist","Data Engineer","ML Engineer",
"Business Analyst"]
counts = [300, 500, 450, 200, 150]
plt.bar(roles, counts)
plt.title("Distribution of Data Science Roles")
plt.xlabel("Roles")
plt.ylabel("Counts")
plt.show()
```

SAMPLE OUTPUT:



RESULT :

Visualization of various Data Science Roles is done using the Bar Graph

Exp No:1.c Conduct an experiment to differentiate Structured,Un-structured and Semi structured data based on data set given.

Description: Create small datasets for each type and explain their characteristics.

CODE:

```
import pandas as pd
structured_data = pd.DataFrame({
'ID': [1, 2, 3],
'Name': ['Alice', 'Bob', 'Charlie'],
'Age': [25, 30, 35]
})
print("Structured Data:\n", structured_data)

unstructured_data = "This is an example of unstructured data. It can be a
piece of text, an image, or a video file."
print("\nUnstructured Data:\n", unstructured_data)

semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height':
165, 'Weight': 68}}
print("\nSemi-structured Data:\n", semi_structured_data)
```

SAMPLE OUTPUT:

```
Structured Data:
    ID     Name  Age
0   1    Alice   25
1   2      Bob   30
2   3  Charlie   35

Unstructured Data:
 This is an example of unstructured data. It can be a piece of text, an
image, or a video file.

Semi-structured Data:
 {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
```

Exp No:1.d Conduct an experiment to encrypt and decrypt given sensitive data.

Description: Use the cryptography library to encrypt and decrypt a piece of data.

CODE:

```
from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Rajalakshmi Engineering College")
token
b'...'
f.decrypt(token)
b'Rajalakshmi Engineering College'
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Rajalakshmi Engineering College."
cipher_text = cipher_suite.encrypt(plain_text)

decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

SAMPLE OUTPUT:

```
Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data:
b'gAAAAABqhGYGvd0sDOcfbFk4sdY_6h6_2d3MtqKc0kQjhMJZYZ0PrN3UJOSoR76V-JHsNLaU
zmdgRlcmC3u7G-VnAyc3AxIir017L8PGIObqpJ1A5ePgss5UjWg7k0u1Ww0RLGn-rbRN'
Decrypted Data: b'Rajalakshmi Engineering College.'
```

RESULT:

The Encryption and Decryption of given dataset is done successfully.